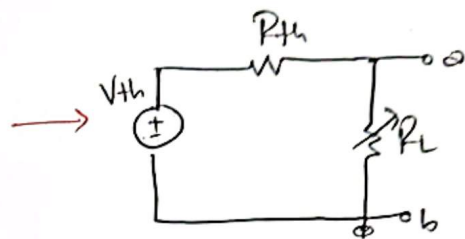
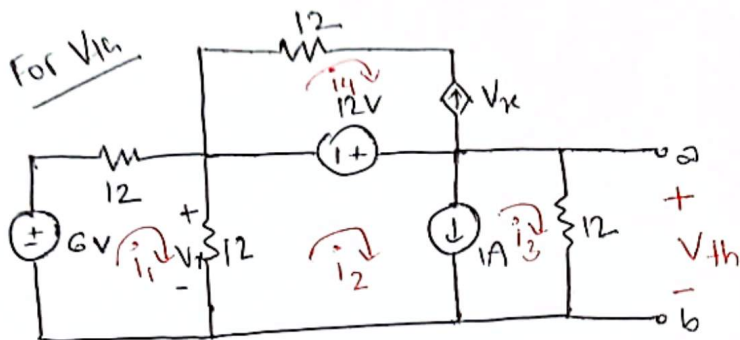


set A (spring 20)

① For V_{th}



mesh 1: $-6 + 12i_1 + 12(i_1 - i_2) = 0$

$\Rightarrow 24i_1 - 12i_2 = 6$ — (i)

mesh 2, 3: $12(i_2 - i_1) - 12 + 12i_3 = 0$

$\Rightarrow -12i_1 + 12i_2 + 12i_3 = 12$ — (ii)

KCL: $i_2 - i_3 = 1$ — (iii)

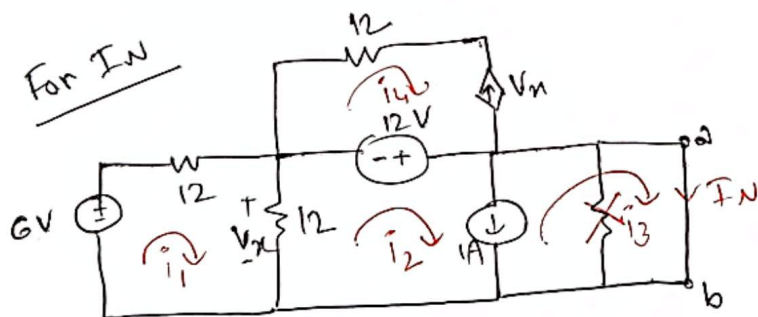
solving, $i_1 = 1A$

$\therefore V_{th} = 12i_3 = 6V$

$i_2 = 1.5A$

$i_3 = 0.5A$

For I_N



mesh 1: $-6 + 12i_1 + 12(i_1 - i_2) = 0$

$\Rightarrow 24i_1 - 12i_2 = 6$ — (i)

mesh 2, 3: $12(i_2 - i_1) - 12 = 0$

$\Rightarrow -12i_1 + 12i_2 = 12$ — (ii)

solving, $i_1 = 1.5A$, $i_2 = 2.5A$, $i_3 = 1.5A$

KCL: $i_2 - i_3 = 1$ — (iii)

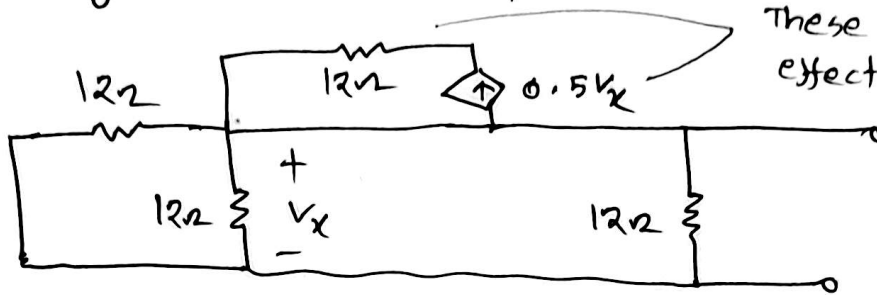
$\therefore I_N = i_3 = 1.5A$

$R_{th} = \frac{V_{th}}{I_N} = \frac{6}{1.5} = 4\Omega$

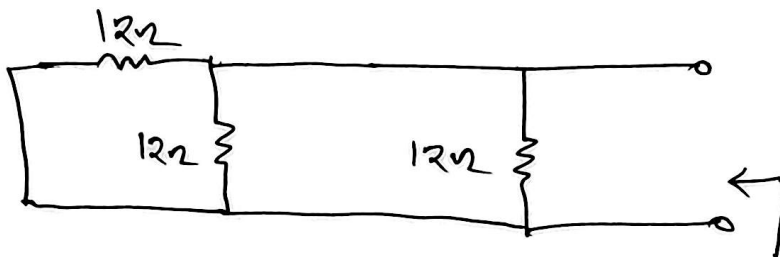
$\therefore P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{6^2}{4 \times 4} = 2.25W$

Another method for $R_{Th} = R_N$

Turning off the independent sources,



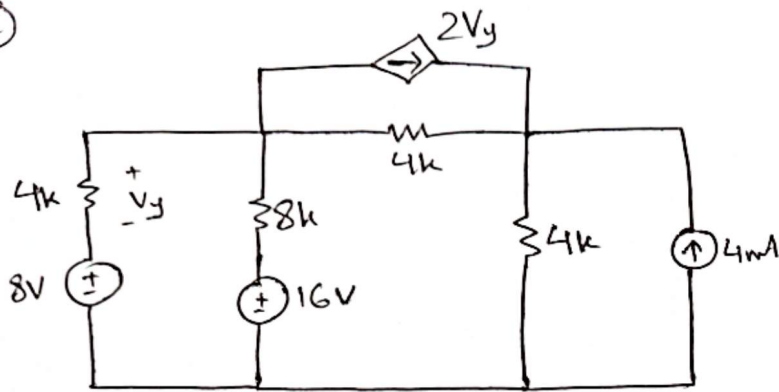
These two don't have any effect on the other part of the circuit because of the short circuit



$$R_{Th} = R_N = 12 \parallel 12 \parallel 12 = 4\Omega$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = 2.25 \text{ W}$$

②



From single loop,

KVL:

$$0 = -8 + 4i + 4i + 16 - 4V_y$$

①

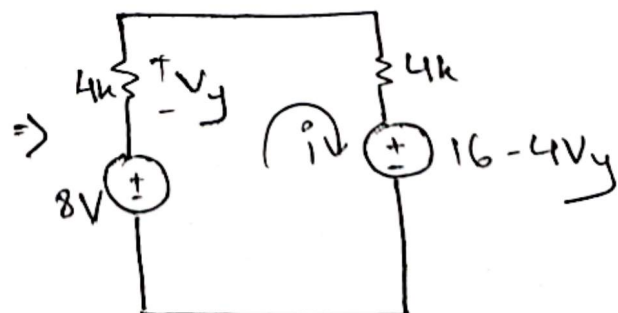
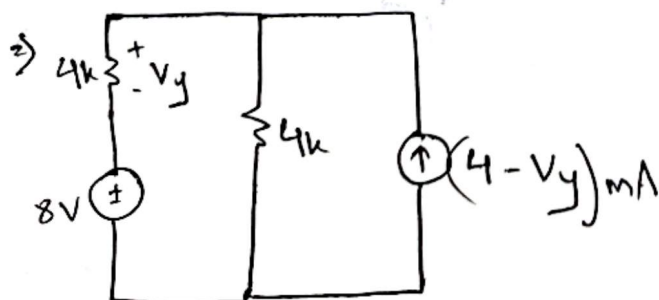
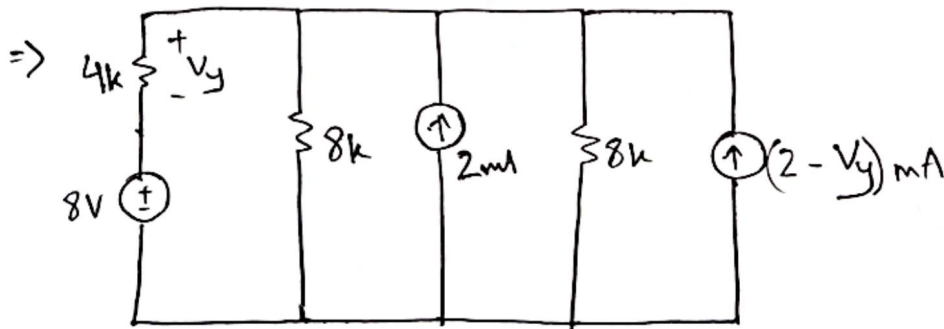
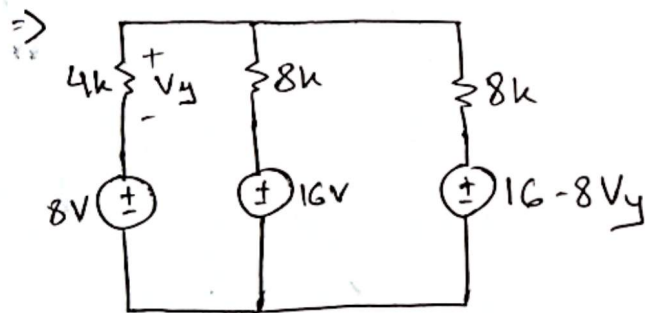
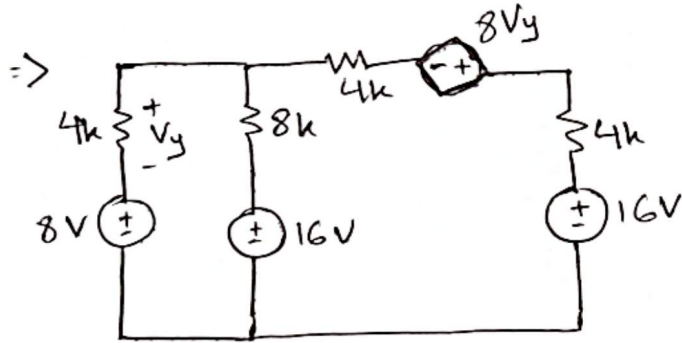
$$V_y = -4i - ②$$

From ①,

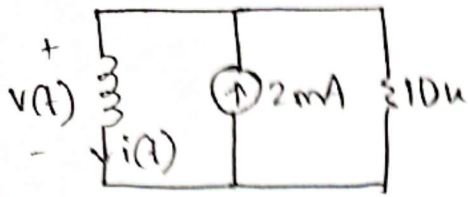
$$-8 - V_y - V_y + 16 - 4V_y = 0$$

$$\Rightarrow -6V_y = -8$$

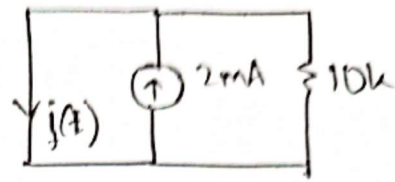
$$V_y = +4/3 \text{ V}$$



③ ∞ At position ∞, ($t < 0$)

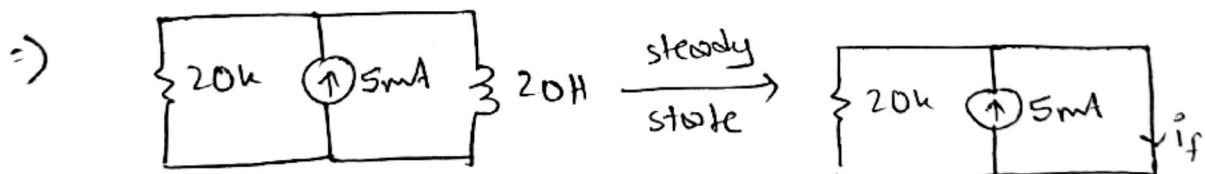
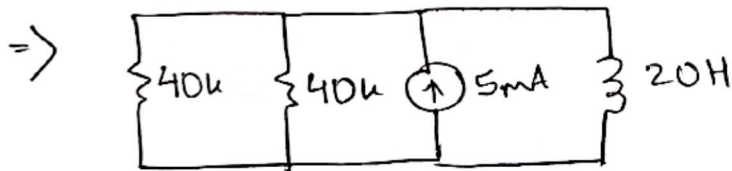
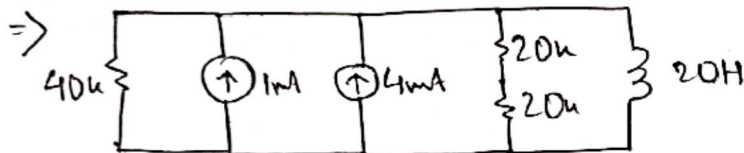
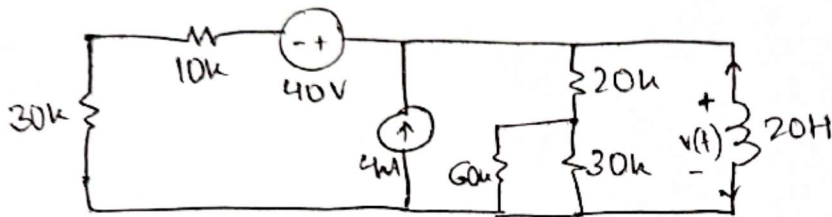


steadily
state



$$i_f = 2 \text{ mA} \longrightarrow i_o \text{ for } t > 0$$

At position b, ($t > 0$)



steadily
state

$$\text{Here, } \tau = L/R = 20/20k = 1 \text{ ms}$$

$$i_f = 5 \text{ mA}$$

$$\begin{aligned} \therefore i(t) &= i_f + (i_o - i_f) e^{-t/\tau} \\ &= 5 + (2 - 5) e^{-t/1 \text{ ms}} \\ &= 5 - 3e^{-t/1 \text{ ms}} \text{ mA} \end{aligned}$$

⑥ $V = L \frac{di}{dt}$

$$\begin{aligned} &= 20 \times \frac{d}{dt} (5 - 3e^{-t/1 \text{ ms}}) \times 10^{-3} \\ &= 20 \times 3 \times \frac{1}{1 \text{ ms}} e^{-t/1 \text{ ms}} \times 10^{-3} \\ &= 60 e^{-t/1 \text{ ms}} \text{ V} \end{aligned}$$

④ ② $V_i = 4$, $V_f = 24$

$$V_c(t) = 24 + (4 - 24)e^{-t/\tau}$$

$$= 24 - 20e^{-t/\tau}$$

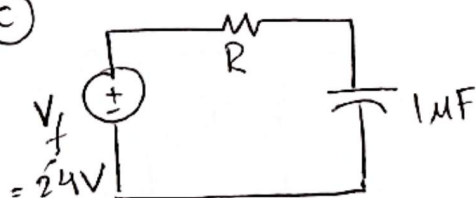
⑥ $i_c(t) = C \frac{dV_c}{dt}$

$$= 1 \mu F \times \frac{d}{dt} (24 - 20e^{-t/\tau})$$

$$= 1 \mu F \times \frac{20}{\tau} e^{-t/\tau}$$

$$= \frac{20 \times 10^{-6}}{\tau} e^{-t/\tau}$$

③



From graph, at $t = 0$,

$$i_c(0) = 10 \text{ mA}$$

$$\Rightarrow \frac{20 \times 10^{-6}}{\tau} e^0 = 10 \text{ mA}$$

$$\Rightarrow \tau = 2 \times 10^{-3} \text{ s} = 2 \text{ ms}$$

⑤ ② $W = \frac{1}{2} CV_s^2$

$$\Rightarrow 25 \text{ mJ} = \frac{1}{2} \times 2 \times 10^{-3} \times V_s^2$$

$$V_s = 5 \text{ V}$$

⑥ $16 \text{ mJ} = \frac{1}{2} \times 2 \times 10^{-3} \times V^2$

$$V = 4 \text{ V (safe)}$$

[③ continued]

For $V_c(t) = 4$,

$$\Rightarrow 5 - 5e^{-t/5} = 4$$

$$\Rightarrow 1/5 = e^{-t/5}$$

$$t = -5 \ln(1/5) = 8.047 \text{ s}$$

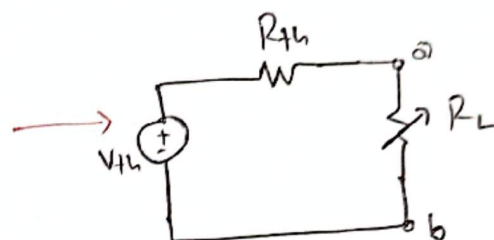
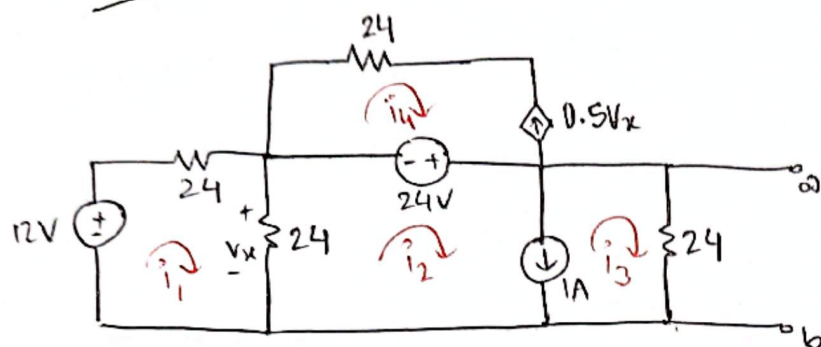
② $\tau = RC = 1k \times 2mF = 2s$

③ $V_f = 5V$, $V_i = 0V$, $\tau = RC = 2.5k \times 2mF = 5s$ $5\tau = 5 \times 2 = 10s$

$$V_c(t) = 5 - 5e^{-t/5}$$

set B (Spring 26)

① For V_{th}



$$\text{mesh 1: } -12 + 24i_1 + 24(i_1 - i_2) = 0$$

$$\Rightarrow 48i_1 - 24i_2 = 12 \quad \text{--- (i)}$$

$$\text{mesh 2, 3: } 24(i_2 - i_3) - 24 + 24i_3 = 0$$

$$\Rightarrow -24i_1 + 24i_2 + 24i_3 = 24 \quad \text{--- (ii)}$$

$$\text{KCL: } i_2 - i_3 = 1 \quad \text{--- (iii)}$$

$$\text{solving, } i_1 = 1 \text{ A}$$

$$i_2 = 1.5 \text{ A}$$

$$i_3 = 0.5 \text{ A}$$

$$V_{th} = 24i_3 = 12 \text{ V}$$

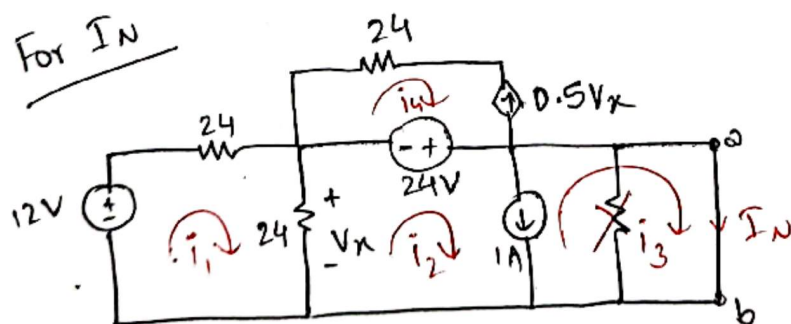
$$V_x = 24(i_1 - i_2)$$

$$= -12 \text{ V}$$

$$i_4 = -V_x$$

$$= +12 \text{ V}$$

For I_N



$$\text{mesh 1: } -12 + 24i_1 + 24(i_1 - i_2) = 0$$

$$48i_1 - 24i_2 = 12 \quad \text{--- (i)}$$

$$\text{mesh 2, 3: } 24(i_2 - i_1) - 24 = 0$$

$$-24i_1 + 24i_2 = 24 \quad \text{--- (ii)}$$

$$\text{KCL: } i_2 - i_3 = 1 \quad \text{--- (iii)}$$

$$\text{solving, } i_1 = 1.5 \text{ A}$$

$$i_2 = 2.5 \text{ A}$$

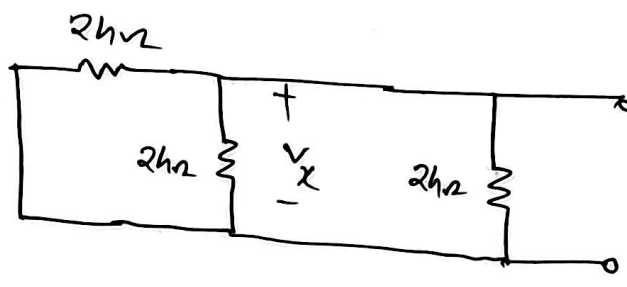
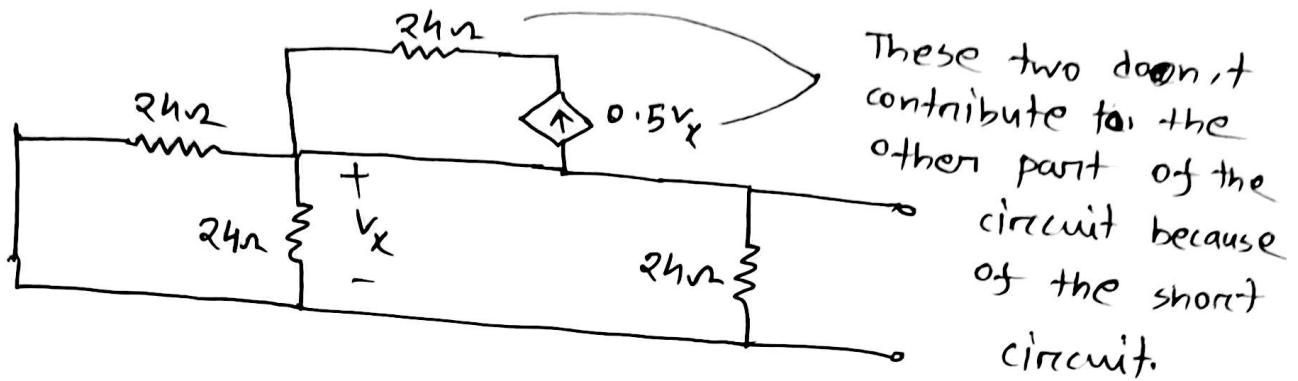
$$i_3 = 1.5 \text{ A} = I_N$$

$$\therefore R_{th} = V_{th} / I_N = 8 \Omega$$

$$\therefore P_{max} = \frac{V_{th}^2}{4R_{th}} = 4.5 \text{ W}$$

Another method for $R_{Th} = R_N$,

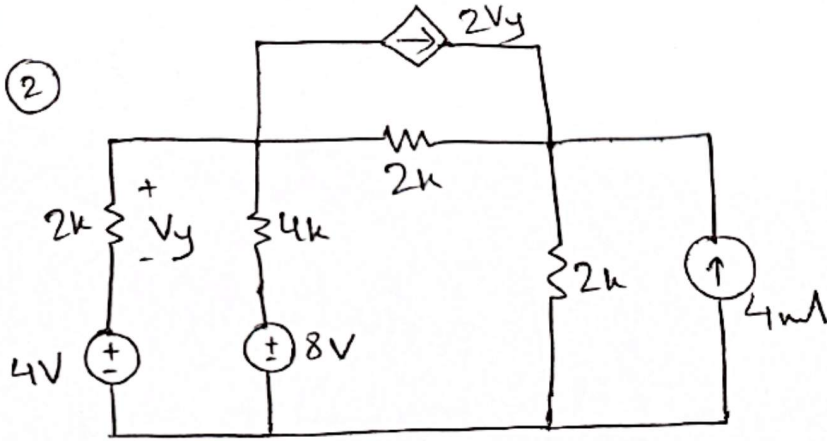
Turning off the independent sources,



$$R_{Th} = 24 \parallel 24 \parallel 24$$

$$= 8\Omega$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = 4.5W$$



From single loop,

KVL:

$$-4 + 2i + 2i + 8 - 2V_y = 0$$

(i)

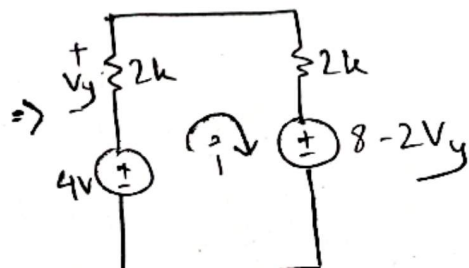
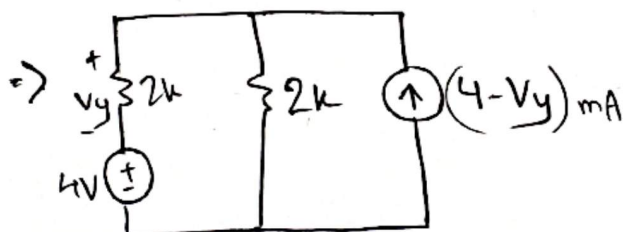
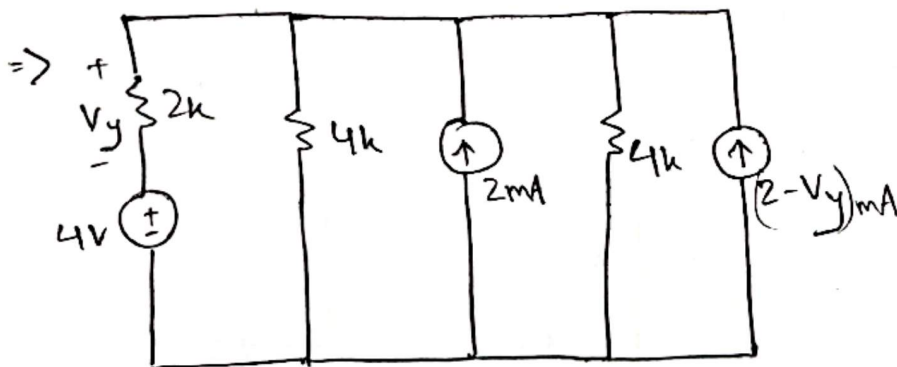
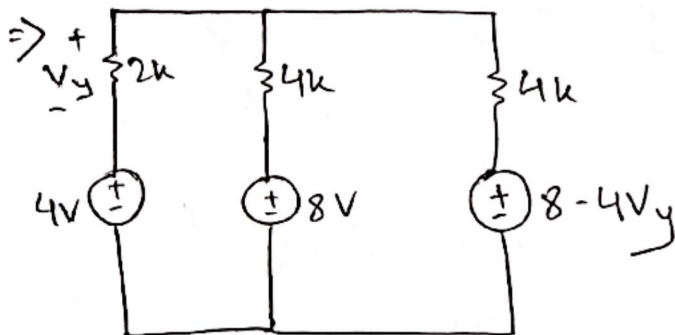
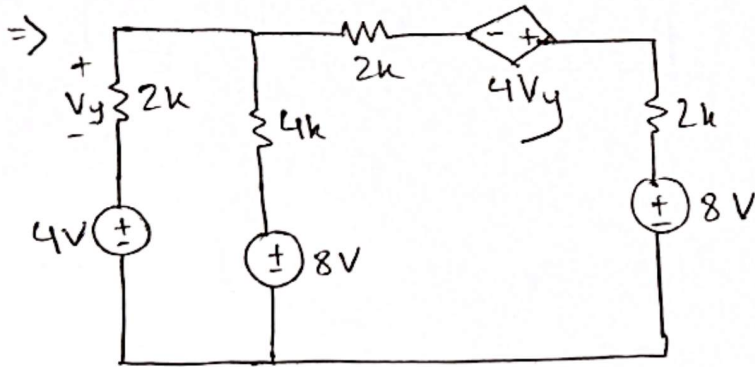
$$V_y = -2i - \textcircled{ii}$$

From (i),

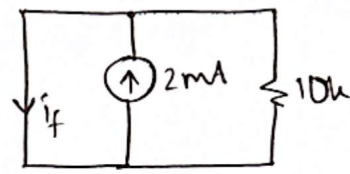
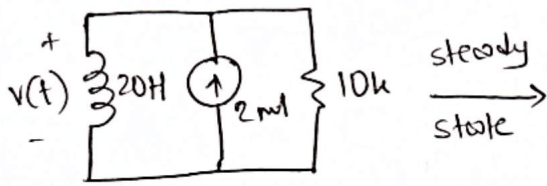
$$-4 - V_y - V_y + 8 - 2V_y = 0$$

$$\Rightarrow -4V_y = -4$$

$$V_y = +1 \text{ V}$$

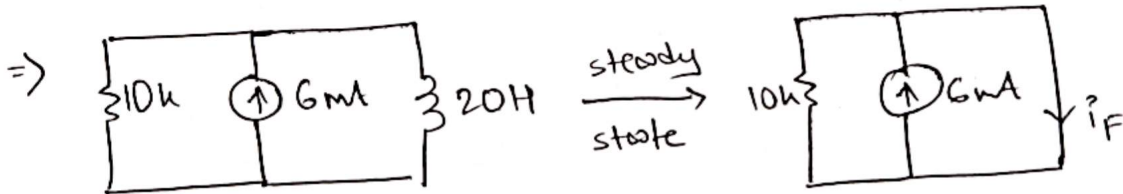
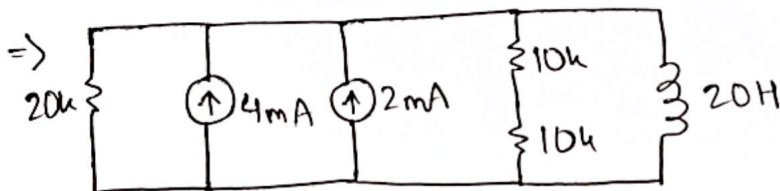
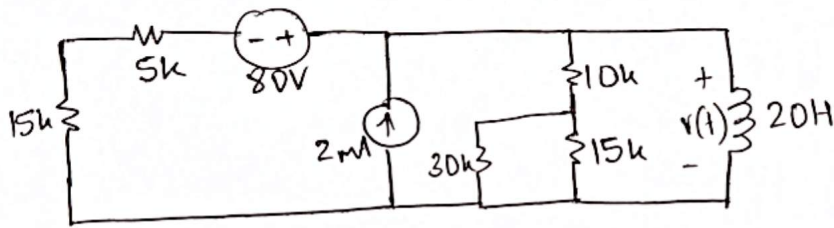


③ At position a, ($t < 0$)



$$\hat{i}_f = 2\text{mA} \rightarrow i_o \text{ for } t > 0$$

At position b, ($t > 0$)



$$\text{Here, } \tau = L/R = 20/10k = 2\text{ms}$$

$$\begin{aligned} i(t) &= \hat{i}_f + (i_o - \hat{i}_f)e^{-t/\tau} \\ &= 6 + (2 - 6)e^{-t/2\text{ms}} \\ &= 6 - 4e^{-t/2\text{ms}} \text{ mA} \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad v &= L \frac{di}{dt} \\ &= 20 \times \frac{d}{dt} (6 - 4e^{-t/2\text{ms}}) \times 10^{-3} \\ &= 20 \times -4 \times -\frac{1}{2\text{ms}} e^{-t/2\text{ms}} \times 10^{-3} \\ &= 40e^{-t/2\text{ms}} \text{ V} \end{aligned}$$

$$(4) \textcircled{\infty} V_i = 2, V_f = 12$$

$$V_c(t) = 12 + (2-12)e^{-t/\tau}$$

$$= 12 - 10e^{-t/\tau}$$

(c) From graph, at $t=0$,

$$i_c(0) = 5 \text{ mA}$$

$$\Rightarrow \frac{10 \times 10^{-6}}{\tau} e^0 = 5 \text{ mA}$$

$$\Rightarrow \tau = 2 \text{ ms}$$

$$(b) i_c(t) = C \frac{dV}{dt}$$

$$= 1 \mu\text{F} \times \frac{d}{dt} (12 - 10e^{-t/\tau})$$

$$= 1 \mu\text{F} \times \frac{10}{\tau} e^{-t/\tau}$$

$$= \frac{10 \times 10^{-6}}{\tau} e^{-t/\tau}$$

$$(5) \textcircled{\infty} W = \frac{1}{2} C V_s^2$$

$$\Rightarrow 25 \text{ mJ} = \frac{1}{2} \times 2 \text{ mF} \times V_s^2$$

$$V_s = 5 \text{ V}$$

$$(b) 9 \text{ mJ} = \frac{1}{2} \times 2 \text{ mF} \times V_s^2$$

$$V = 3 \text{ V (scale)}$$

$$(c) V_i = 0, V_f = 5 \text{ V}, \tau = RC = 3 \text{ k} \times 2 \text{ mF}$$

$$= 6 \text{ s}$$

$$V_c(t) = 5 - 5e^{-t/6}$$

$$\text{For } V_c(t) = 3,$$

$$\Rightarrow 5 - 5e^{-t/6} = 3$$

$$\Rightarrow \frac{2}{5} = e^{-t/6}$$

$$t = -6 \ln\left(\frac{2}{5}\right) = 5.498 \text{ s}$$

$$(a) \tau = RC = 2 \text{ k} \times 2 \text{ mF}$$

$$= 4 \text{ s}$$

$$5\tau = 5 \times 4 = 20 \text{ s}$$